

Madhav Mishra

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech in Computer Science and Engineering (Cyber Security)

Oct. 2022 – July 2027

CGPA:8.12/10

EXPERIENCE

Chhattisgarh Infotech Promotion Society (Govt. of Chhattisgarh)

Cyber Security Intern

Oct 2025 – Dec 2025

Raipur, Chhattisgarh

- * Observed and assisted in day-to-day cybersecurity activities within the organization.
- * Learned basic concepts of vulnerability assessment, system security, and network security under the guidance of senior team members.
- * Prepared documentation and reports related to assigned cybersecurity tasks and security best practices.

PROJECTS

Secure Automated Grading System | *Python, JavaScript, Golang*

April 2025

- * Designed a backend system to automate evaluation of student submissions with role-based access control.
- * Implemented secure input handling and validation to reduce common web vulnerabilities (SQLi, command injection).
- * Integrated basic logging and error monitoring to support incident investigation and debugging.

Hybrid Dual-Signature Scheme (HDSS) | *Python, Golang*

Feb 2025 – Oct 2025

- * Designed and implemented a hybrid post-quantum digital signature model combining Falcon and SPHINCS+ to improve resilience against quantum attacks in blockchain and IoT scenarios.
- * Built a proof-of-concept benchmarking workflow to compare HDSS with RSA, ECDSA, and individual PQC schemes using real blockchain and IoT datasets, evaluating signature validity, size, and latency.
- * Analyzed security-performance trade-offs of hybrid signatures, demonstrating 100% tamper detection in preliminary experiments while documenting deployment limitations and future optimization scope.

TECHNICAL SKILLS

- * **Languages:** Python, Golang, Bash
- * **Frameworks & Libraries:** Flask, Django, NumPy, Gin, Gorilla Mux, Crypto, Pycrypto
- * **Security Tools:** Wireshark, Nmap, Burp Suite, Metasploit, Kali Linux, OWASP ZAP, Nikto, BEEF, GOBuster.

ACHIEVEMENT

National Finalist, SIH

- * Developed scalable real-time bus scheduling and route optimization system.
- * Integrated GPS tracking, geospatial APIs, and automated alert mechanisms.

Represented college Cricket team at Advitya annual fest

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02) Amazon Web Services (AWS), 2026

PUBLICATIONS

Madhav Mishra, “Hybrid Dual-Signature Scheme using Falcon and SPHINCS+: A Theoretical Study with Preliminary Results,” *IEEE [ISCS]*, 2025.

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